Part 1. Getting started with blockchain programming

Blockchain is poised to become an integral part of existing computing systems as a trust layer. So part 1 begins with an overview of a blockchain as decentralized infrastructure, disintermediator, and distributed ledger technology: the three Ds of blockchain. The three Ds together enable trust in decentralized applications with the help of an essential coding element called smart contracts. Part 1 focuses on the design and development of smart contracts in incremental steps. Design diagrams and design principles are introduced to guide the smart contract design. You'll learn to code smart contracts by using the Solidity language, and deploy and test them by using a web-based Remix IDE. Then you'll learn to code rules for trust and integrity in the smart contracts. Finally I introduce the Truffle suite of tools with detailed instructions on installing it. You'll be using Truffle commands throughout the book for deploying and testing smart contracts.

Chapter 1 is about blockchain basics. Chapter 2 introduces essential coding elements of a smart contract, using a counter (Counter.sol) application and an airline consortium (ASK.sol) for trading seats. Chapter 3 illustrates techniques for verification and validation of the smart contract, using a digital democracy application. Chapter 4 demonstrates the